

BLACK HILLS, SD, USA

WMO: 726605

Lat: 44.483N

Lon: 103.783W

Elev: 3931

StdP: 12.72

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 00386

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.1	-2.3	-15.2	2.8	-6.0	-9.3	3.9	-1.2	32.4	47.0	28.2	44.2	6.3	300	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	90.6	65.3	87.7	64.9	84.1	64.7	70.7	82.0	68.8	80.7	66.9	79.7	10.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	112.5	78.7	64.2	104.1	75.4	63.1	100.2	73.8	37.3	82.2	35.4	80.1	33.9	79.9	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.9	24.1	20.7	DB	-14.8	96.0	3.8	4.1	-17.5	99.0	-19.8	101.4	-21.9	103.7	-24.7	106.7	(4)
(5)			WB	-15.1	74.4	3.6	2.3	-17.7	76.1	-19.8	77.4	-21.9	78.7	-24.5	80.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.2	27.1	24.5	36.4	42.9	51.7	62.9	70.6	68.1	61.0	46.3	36.2	25.6		
	DBStd	19.18	12.82	14.50	12.56	10.01	8.71	7.17	5.94	6.14	8.84	10.61	12.01	13.87		
	HDD50	3603	711	717	442	249	82	4	0	0	18	194	428	758		
	HDD65	7294	1175	1134	887	663	420	118	14	39	175	585	863	1221		
	CDD50	2231	2	4	20	36	137	392	639	560	347	78	14	2		
	CDD65	445	0	0	0	1	9	56	188	134	53	4	0	0		
	CDH74	5321	0	0	4	22	110	790	2084	1535	713	63	0	0		
	CDH80	1865	0	0	0	2	17	244	814	562	215	11	0	0		
	WSAvg	9.1	10.6	9.4	9.3	10.0	9.3	8.7	7.8	7.6	7.8	9.4	9.4	9.6		
	PrecAvg	25.3	1.1	1.2	1.7	2.8	4.0	3.8	2.6	1.9	1.7	2.3	1.3	1.0		
(15) Precipitation	PrecMax	36.1	2.4	2.2	3.3	6.3	9.9	8.3	5.6	3.9	5.0	7.1	3.9	2.4		
	PrecMin	17.5	0.4	0.4	0.9	0.9	1.3	0.7	0.7	0.6	0.7	0.1	0.1	0.1		
	PrecStd	5.2	0.5	0.4	0.6	1.2	2.2	2.1	1.3	1.0	1.1	1.9	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	63.8	73.0	77.1	82.1	91.4	96.6	93.3	90.3	81.3	70.1	62.5		
		MCWB	40.9	45.1	50.5	53.9	63.2	66.3	64.9	63.3	63.2	57.5	49.2	44.7		
	2%	DB	53.5	56.6	66.2	70.4	76.8	86.0	91.2	90.0	85.7	74.8	64.1	54.1		
		MCWB	40.2	42.7	47.1	52.1	59.3	64.8	66.1	64.9	62.6	53.3	46.6	40.2		
	5%	DB	48.4	50.2	61.2	64.2	72.7	82.0	88.1	85.9	81.7	68.5	59.1	48.0		
		MCWB	37.8	38.9	44.0	49.1	58.2	64.2	65.9	64.6	60.6	51.5	44.1	37.2		
	10%	DB	45.4	45.1	55.0	59.5	68.0	78.6	84.1	82.3	77.3	63.8	54.3	44.7		
		MCWB	35.8	35.7	42.1	46.5	55.8	63.2	66.2	63.4	58.5	49.3	41.4	35.1		
	0.4%	WB	42.8	46.1	51.5	56.0	67.4	72.0	74.9	71.3	69.1	59.0	50.4	45.3		
		MCDB	56.1	60.8	70.8	72.9	79.4	83.1	85.7	82.8	83.6	76.3	67.7	61.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.5	42.5	47.7	52.7	62.2	68.9	71.6	69.1	65.6	55.7	47.5	40.7		
		MCDB	51.3	55.7	63.9	67.4	73.3	80.1	82.7	81.8	79.7	70.3	63.0	51.8		
	5%	WB	38.0	39.1	45.2	50.4	59.4	66.4	69.4	67.5	62.9	52.8	44.9	37.6		
		MCDB	47.5	49.8	59.7	63.3	69.6	78.3	80.3	80.1	77.4	66.9	58.8	47.1		
	10%	WB	36.0	35.8	42.6	47.9	56.5	64.1	67.6	65.5	60.3	50.2	41.9	35.2		
		MCDB	44.8	43.9	54.4	59.5	66.1	75.8	80.3	78.8	74.3	61.9	52.8	44.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	18.1	19.9	20.3	19.3	21.9	23.4	23.6	23.8	19.5	19.0	16.6		
		MCDBR	22.9	24.1	28.7	29.5	25.8	28.9	27.7	28.4	29.6	27.8	26.7	21.4		
		MCWBR	14.1	13.7	15.2	16.1	14.8	15.0	11.7	13.3	13.8	14.2	14.2	13.0		
		MCDBR	22.0	23.3	27.4	27.5	23.5	26.3	23.6	24.9	26.5	26.2	26.0	20.3		
	5% WB	MCWBR	14.3	13.8	15.1	15.8	14.3	15.8	14.4	14.2	13.9	14.1	14.1	13.1		
(40) Clear-Sky Solar Irradiance	taub		0.253	0.260	0.277	0.306	0.320	0.334	0.354	0.356	0.320	0.289	0.262	0.249		
	taud		2.583	2.555	2.561	2.472	2.503	2.486	2.460	2.432	2.520	2.581	2.570	2.578		
	Ebn at Noon		286	302	309	305	301	296	288	284	288	286	279	277		
	Edh at Noon		20	24	28	33	33	34	34	34	29	23	20	18		
(44) All-Sky Solar Radiation	RadAvg		589	896	1272	1569	1868	2178	2248	1935	1507	981	655	491		
	RadStd		27	51	98	90	188	177	82	76	98	92	56	35		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page